

Load Testing Report Of FlexEm

Table Of Contents

Overview.....	3
Objective.....	3
Load Test Configuration:.....	3
Response Rate.....	3
Time Intervals.....	3
System Configuration according to Production Server:.....	3
Data Collection:.....	3
Test Under Sixty (60) Minutes.....	4
User Registration API:.....	4
1.1 Test with 09 RPS and 350 User.....	4
1.1.1 Execution Summary.....	4
1.1.2 Observation Summary.....	4
1.1.3 Details.....	4
1.1.4 Summary Report.....	5
1.1.5 Configuration:.....	5
1.1.6 Output Graph.....	6
Aggregate Graph:.....	6
1.1.7 Response Time Over Time.....	6
Response Time Over Time Graph:.....	6
1.2 Test with 09 RPS and 500 User.....	7
1.2.1 Execution Summary.....	7
1.2.2 Observation Summary.....	7
1.2.3 Details.....	7
1.2.4 Summary Report.....	8
1.2.5 Configuration:.....	8
1.2.6 Output Graph.....	9
Aggregate Graph:.....	9
1.2.7 Response Time Over Time.....	9
Response Time Over Time Graph:.....	9
1.3.1 Execution Summary.....	10
1.3.2 Observation Summary.....	10
1.3.3 Details.....	10
1.3.4 Summary Report.....	11
1.3.5 Configuration:.....	11
1.3.6 Output Graph.....	12
Aggregate Graph:.....	12
Response Time Over Time Graph:.....	12
Test Under Sixty (60) MINUTES.....	13
User Login API:.....	13

2.1 Test with 09 RPS and 1000 User.....	13
2.1.1 Execution Summary.....	13
2.1.2 Observation Summary.....	13
2.1.3 Details.....	13
2.1.4 Summary Report.....	14
2.1.5 Configuration:.....	14
2.1.5 Output Graph.....	15
Aggregate Graph:.....	15
Test Under Sixty (60) MINUTES.....	16
Flow -> User Login, Brands By Category, Brands & Offers API:.....	16
3.1 Test with 09 RPS and 1000 User.....	16
3.1.1 Execution Summary.....	16
3.1.2 Observation Summary.....	16
3.1.3 Details.....	16
3.1.4 Summary Report.....	17
3.1.5 Configuration:.....	18
3.1.6 Output Graph.....	18
Aggregate Graph:.....	18
Response Time Over Time Graph:.....	19
Test Under Sixty (60) MINUTES.....	20
Flow -> User Login, Brands By Category, Brands, Offers, Create Claim API:.....	20
4.1 Test with 09 RPS and 900 User.....	20
4.1.1 Execution Summary.....	20
4.1.2 Observation Summary.....	20
4.1.3 Details.....	20
4.1.4 Summary Report.....	21
4.1.5 Configuration:.....	22
4.1.6 Output Graph.....	22
Aggregate Graph:.....	22
Response Time Over Time Graph:.....	23
Test Under Sixty (60) MINUTES.....	24
Flow -> User Login, Brands By Category, Brands, Offers, Create Claim, Get Claim API:....	24
5.1 Test with 09 RPS and 800 User.....	24
5.1.1 Execution Summary.....	24
5.1.2 Observation Summary.....	24
5.1.3 Details.....	24
5.1.4 Summary Report.....	25
5.1.5 Configuration:.....	26
5.1.6 Output Graph.....	26
Aggregate Graph:.....	26
Response Time Over Time Graph:.....	27

Overview

Objective

To evaluate the scalability and responsiveness of the FlexEm system, a structured load testing strategy was designed using JMeter. The primary objective was to assess the system's performance under expected daily usage patterns, simulating real-world user behavior.

According to analytics, FlexEm expects approximately **1000 daily active users (DAUs)**, with user activity concentrated within a **4-hour operational window**. To distribute the load realistically, the expected user count per hour was calculated as:

$$1000 \text{ DAUs} \div 4 \text{ hours} = 250 \text{ active users per hour}$$

This formed the baseline for our load testing design. We ran multiple tests with controlled request rates (e.g., **09 Requests Per Second**) and varying user loads, gradually increasing the users from **350 to 1000** across different API flows. The objective was to analyze the system's behavior under rising demand and identify any potential performance bottlenecks, capacity limits, or failure thresholds.

Load Test Configuration:

Response Rate

- 09 RPS

Time Intervals

- 60 mins

System Configuration according to Production Server:

1. CPU: 1 Core
2. Ram : 2 Gb
3. Database: 400 Mb

Data Collection:

During the load test, various performance metrics have collected, including but not limited to:

- Response times for APIs.
- Throughput and APIs successful response rates.
- Error rates

Test Under Sixty (60) Minutes

User Registration API:

1.1 Test with 09 RPS and 350 User

1.1.1 Execution Summary

Following is the summary of the load test performed with 09 RPS and 350 User under 60 minutes:

Total Requests	350	Error/Request	0/350
Response Time(Avg)	14332 ms	Response Time(Min)	627 ms

1.1.2 Observation Summary

- No fault found till 55 mins
- No error appeared after 60 minutes
- The system is working correctly as we've achieved our anticipated 09 TPS

1.1.3 Details

The following contains the activity report during the load test & also the performance details.

Estimated User Capacity	350
Maximized Users Analyzed	350
Start Time	01/06/2025 12:53:08 PM
End Time	01/06/2025 01:53:05 PM
Duration	60 min
Total Hits	350
Peak hits/ms	28000/ms
Total Failed Hits	0
RPS	9.39
Received KB/sec	9.68

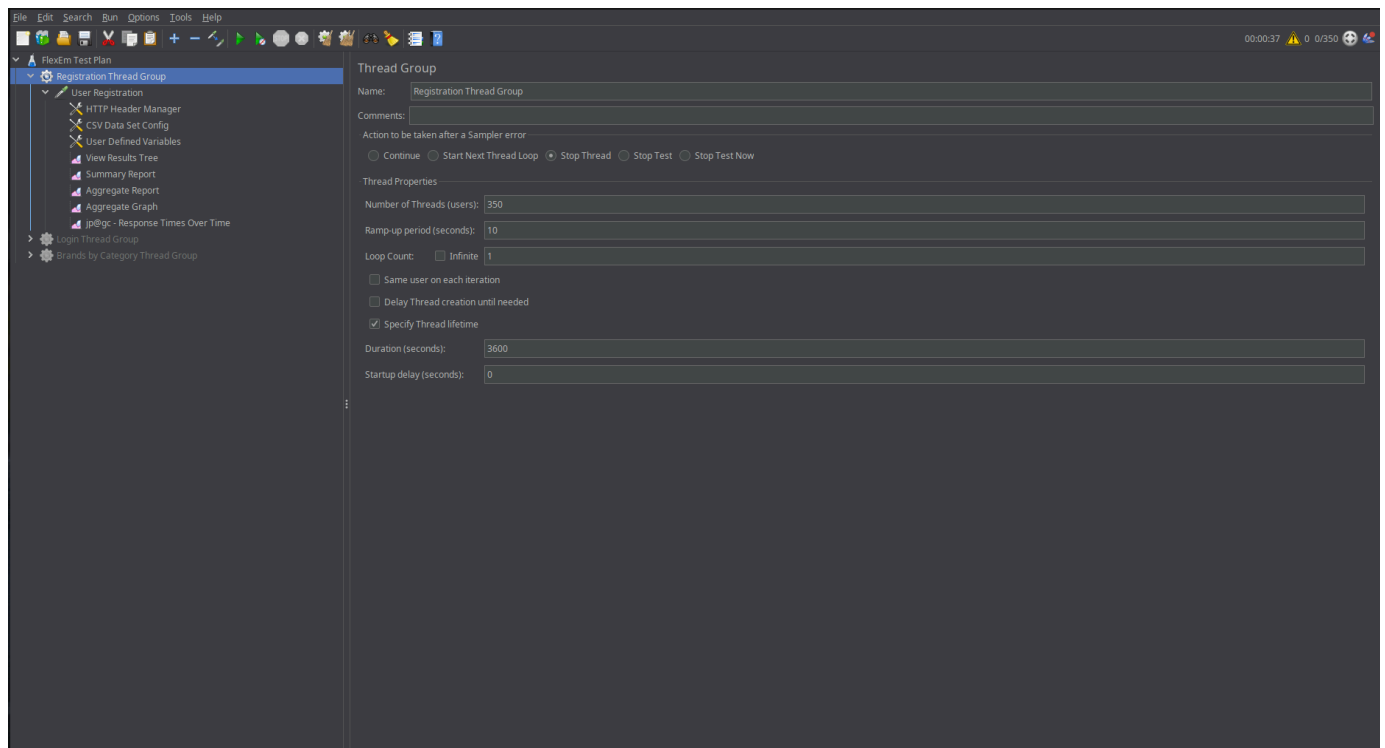
Sent KB/sec	2.94
-------------	------

1.1.4 Summary Report

The summary is the one kind of dashboard perspective on the test while the test is pursued and it has wrapped up. The Summary Report will show up when the test begins to gather Information.

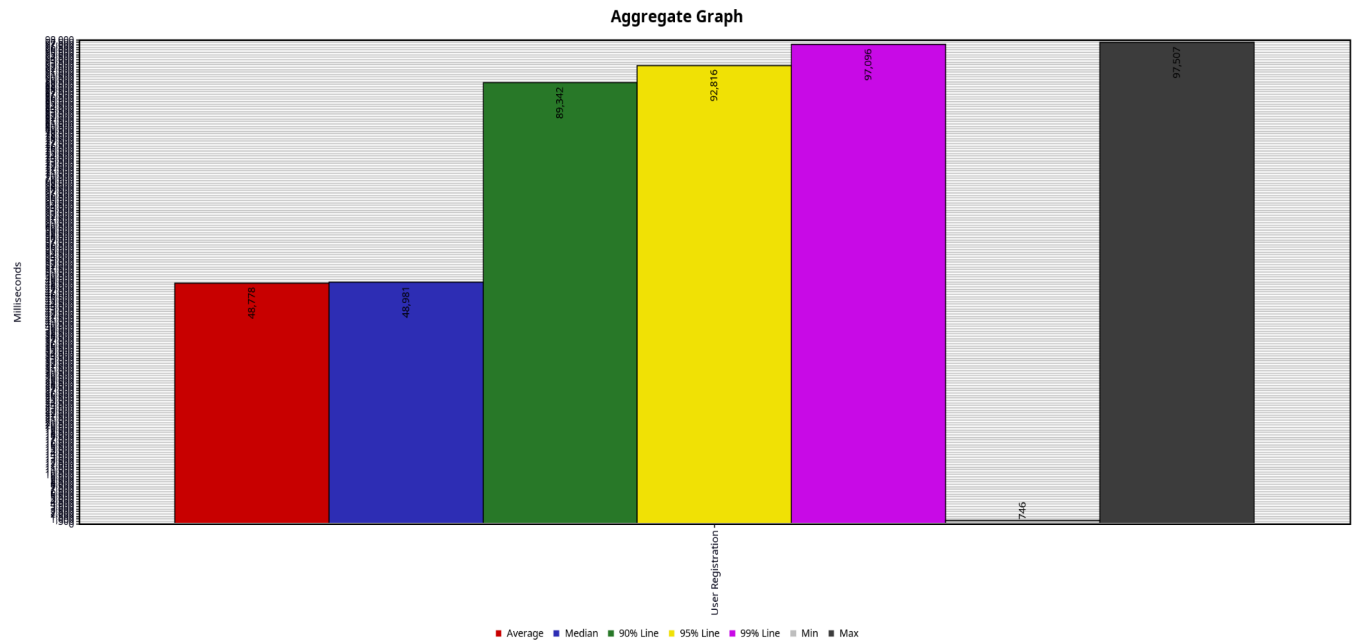
Label	Sample	Average	Min	Max	Std.Dev	Error%	Throughput	Received KB/sec	Sent KB/sec	Avg.Bytes
User Registration	350	14332	627	27294	7821.27	0.000	9.39396	9.68	2.94	1055.2

1.1.5 Configuration:



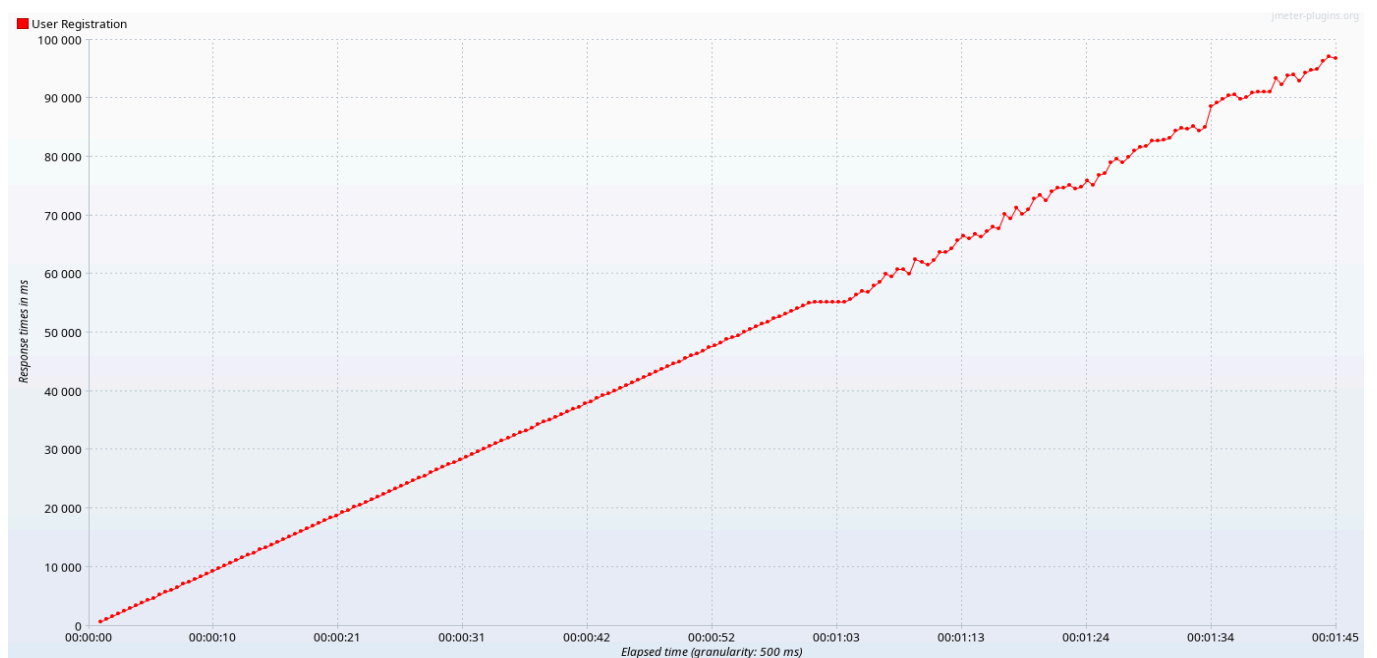
1.1.6 Output Graph

Aggregate Graph:



1.1.7 Response Time Over Time

Response Time Over Time Graph:



1.2 Test with 09 RPS and 500 User

1.2.1 Execution Summary

Following is the summary of the load test performed with 09 RPS and 500 User under 60 minutes:

Total Requests	500	Error/Request	0/500
Response Time(Avg)	22263 ms	Response Time(Min)	626 ms

1.2.2 Observation Summary

- No fault found till 55 mins
- No error appeared after 60 minutes
- The system is working correctly as we've achieved our anticipated 09 TPS

1.2.3 Details

The following contains the activity report during the load test & also the performance details.

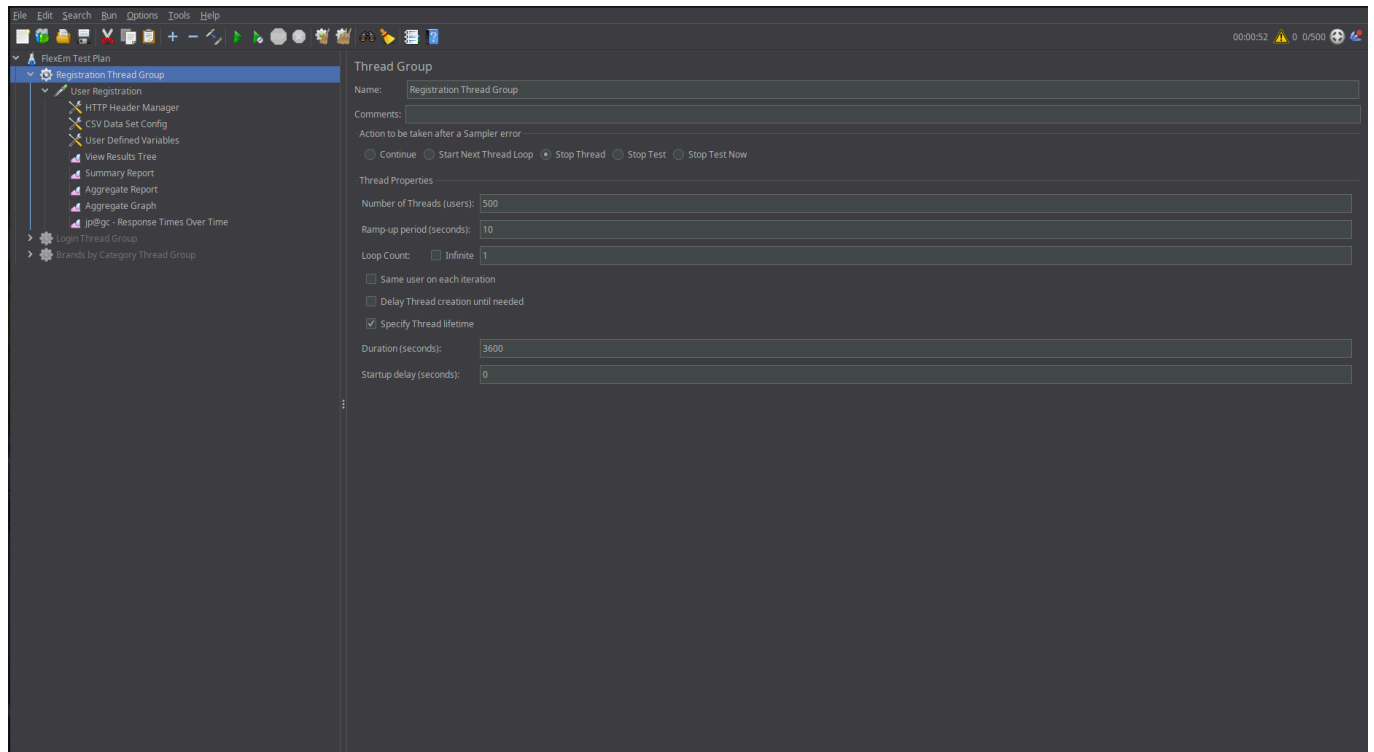
Estimated User Capacity	500
Maximized Users Analyzed	500
Start Time	01/06/2025 01:53:08 PM
End Time	01/06/2025 02:53:05 PM
Duration	60 min
Total Hits	500
Peak hits/ms	43000/ms
Total Failed Hits	0
RPS	9.44
Received KB/sec	9.74
Sent KB/sec	2.96

1.2.4 Summary Report

The summary is the one kind of dashboard perspective on the test while the test is pursued and it has wrapped up. The Summary Report will show up when the test begins to gather Information.

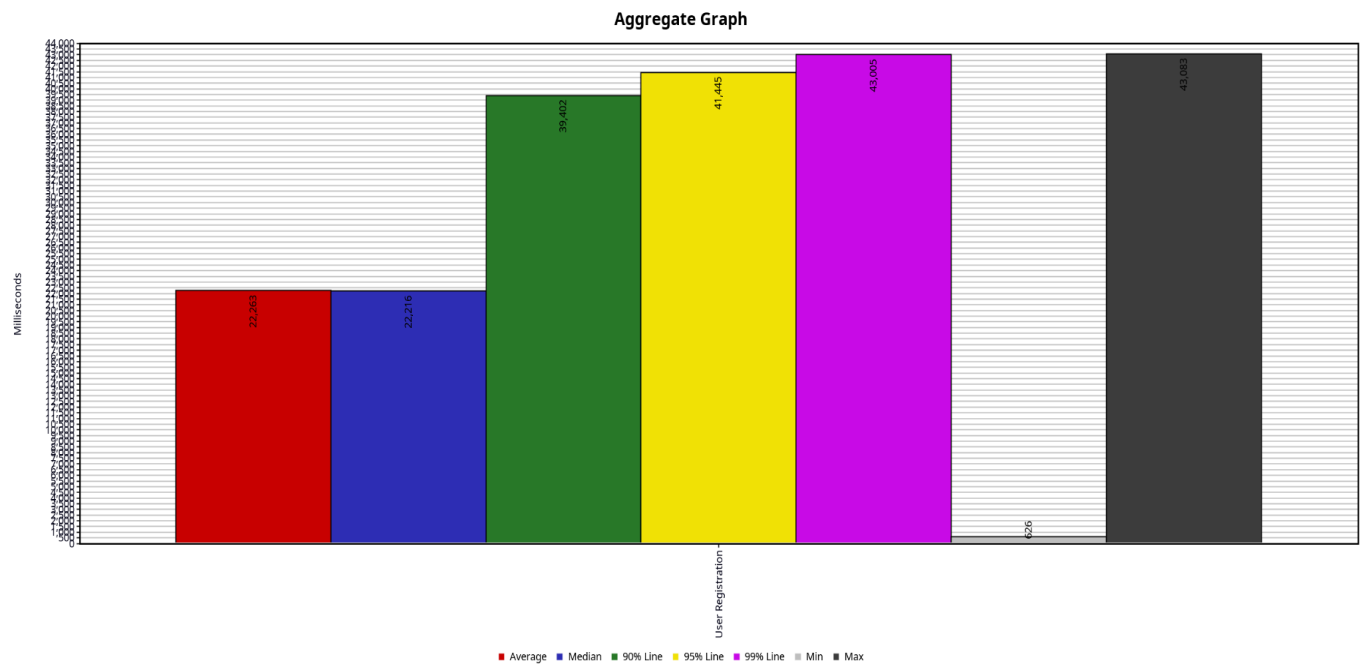
Label	Sample	Average	Min	Max	Std.Dev	Error%	Through put	Received KB/sec	Sent KB/sec	Avg. Bytes
User Registration	500	22263	626	43083	12362.97	0.000	9.44412	9.74	2.96	1056.0

1.2.5 Configuration:



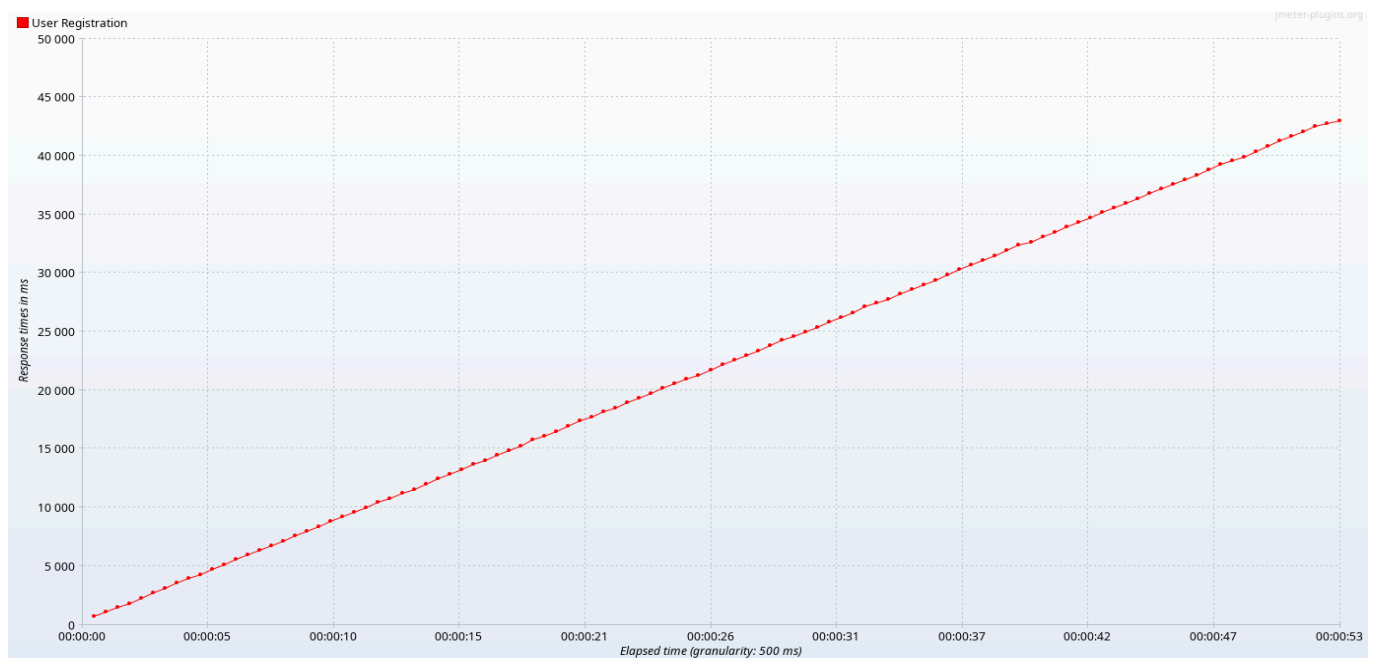
1.2.6 Output Graph

Aggregate Graph:



1.2.7 Response Time Over Time

Response Time Over Time Graph:



1.3 Test with 09 RPS and 1000 User

1.3.1 Execution Summary

Following is the summary of the load test performed with 09 RPS and 1000 User under 60 minutes:

Total Requests	1000	Error/Request	0/1000
Response Time(Avg)	48778 ms	Response Time(Min)	746 ms

1.3.2 Observation Summary

- No fault found till 55 mins
- No error appeared after 60 minutes
- The system is working correctly as we've achieved our anticipated 09 TPS

1.3.3 Details

The following contains the activity report during the load test & also the performance details.

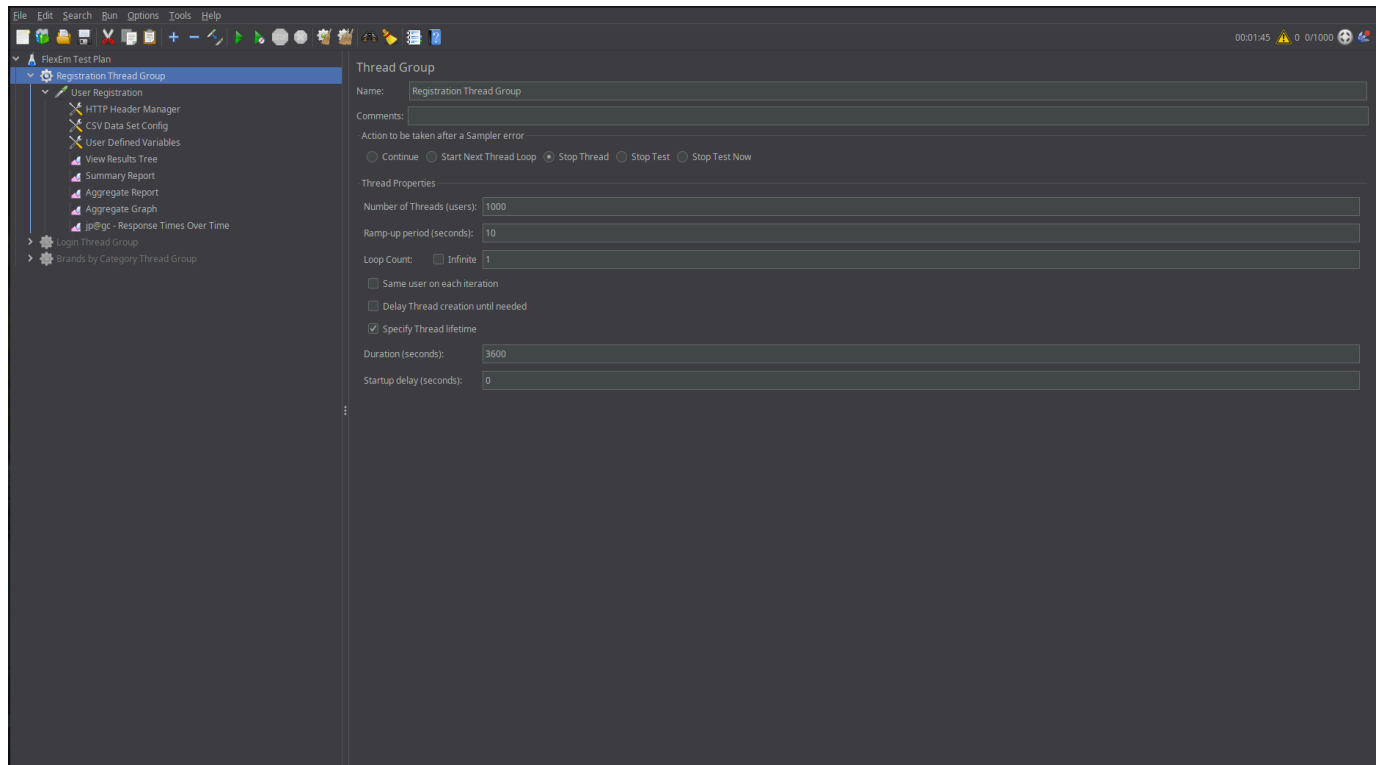
Estimated User Capacity	1000
Maximized Users Analyzed	1000
Start Time	01/06/2025 02:53:08 PM
End Time	01/06/2025 03:53:05 PM
Duration	60 min
Total Hits	1000
Peak hits/ms	95000/ms
Total Failed Hits	0
RPS	9.48
Received KB/sec	9.78
Sent KB/sec	2.97

1.3.4 Summary Report

The summary is the one kind of dashboard perspective on the test while the test is pursued and it has wrapped up. The Summary Report will show up when the test begins to gather Information.

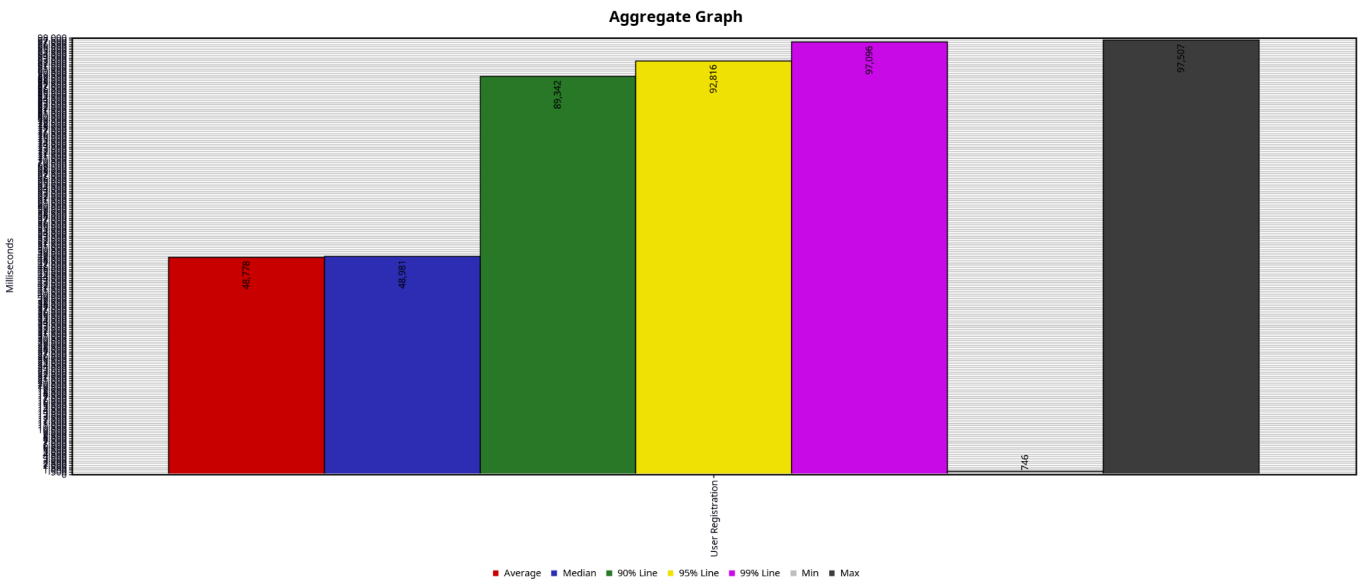
Label	Sample	Average	Min	Max	Std.Dev	Error%	Through put	Received KB/sec	Sent KB/sec	Avg. Bytes
User Registration	1000	48778	746	97507	27775.23	0.000	9.48803	9.78	2.97	1056.0

1.3.5 Configuration:



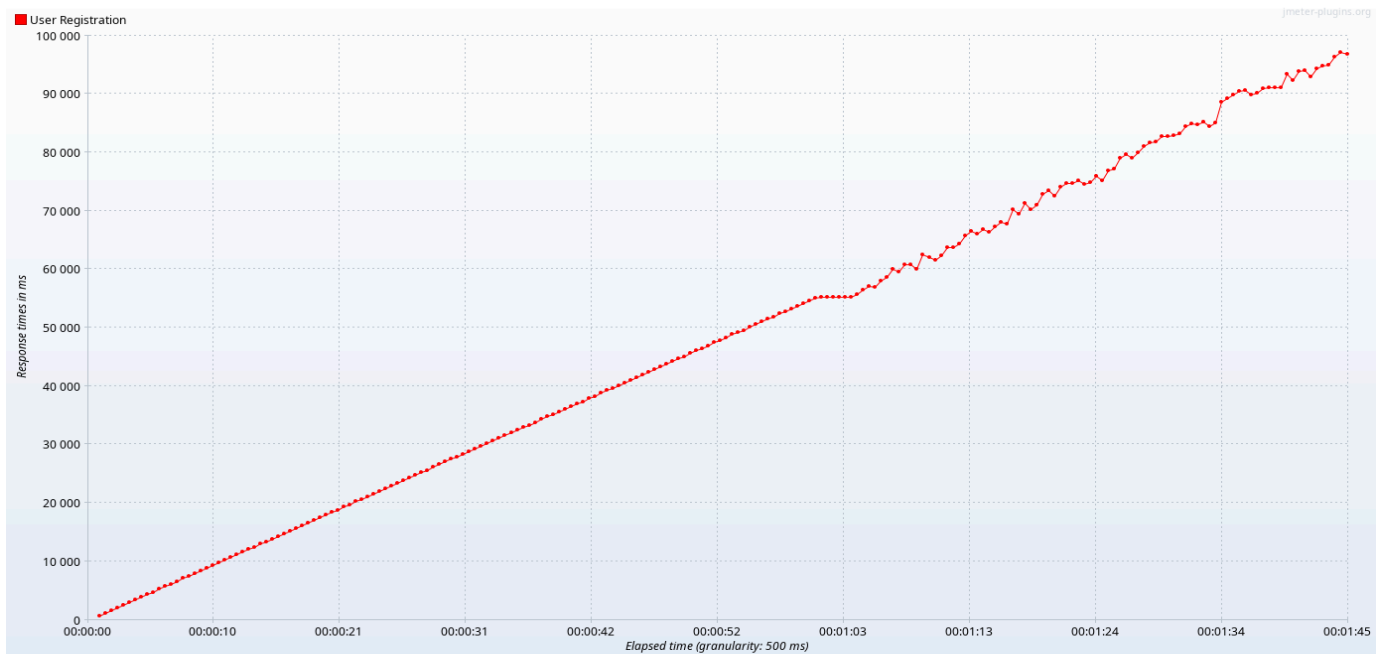
1.3.6 Output Graph

Aggregate Graph:



1.3.7 Response Time Over Time

Response Time Over Time Graph:



Test Under Sixty (60) MINUTES

User Login API:

2.1 Test with 09 RPS and 1000 User

2.1.1 Execution Summary

Following is the summary of the load test performed with 09 RPS and 1000 User under 60 minutes:

Total Requests	1000	Error/Request	0/1000
Response Time(Avg)	47260 ms	Response Time(Min)	107 ms

2.1.2 Observation Summary

- No fault found till 55 mins
- No error appeared after 60 minutes
- The system is working correctly as we've achieved our anticipated 09 TPS

2.1.3 Details

The following contains the activity report during the load test & also the performance details.

Estimated User Capacity	1000
Maximized Users Analyzed	1000
Start Time	01/06/2025 03:53:08 PM
End Time	01/06/2025 04:53:05 PM
Duration	60 min
Total Hits	1000
Peak hits/ms	96000/ms
Total Failed Hits	0
RPS	9.58
Received KB/sec	9.84

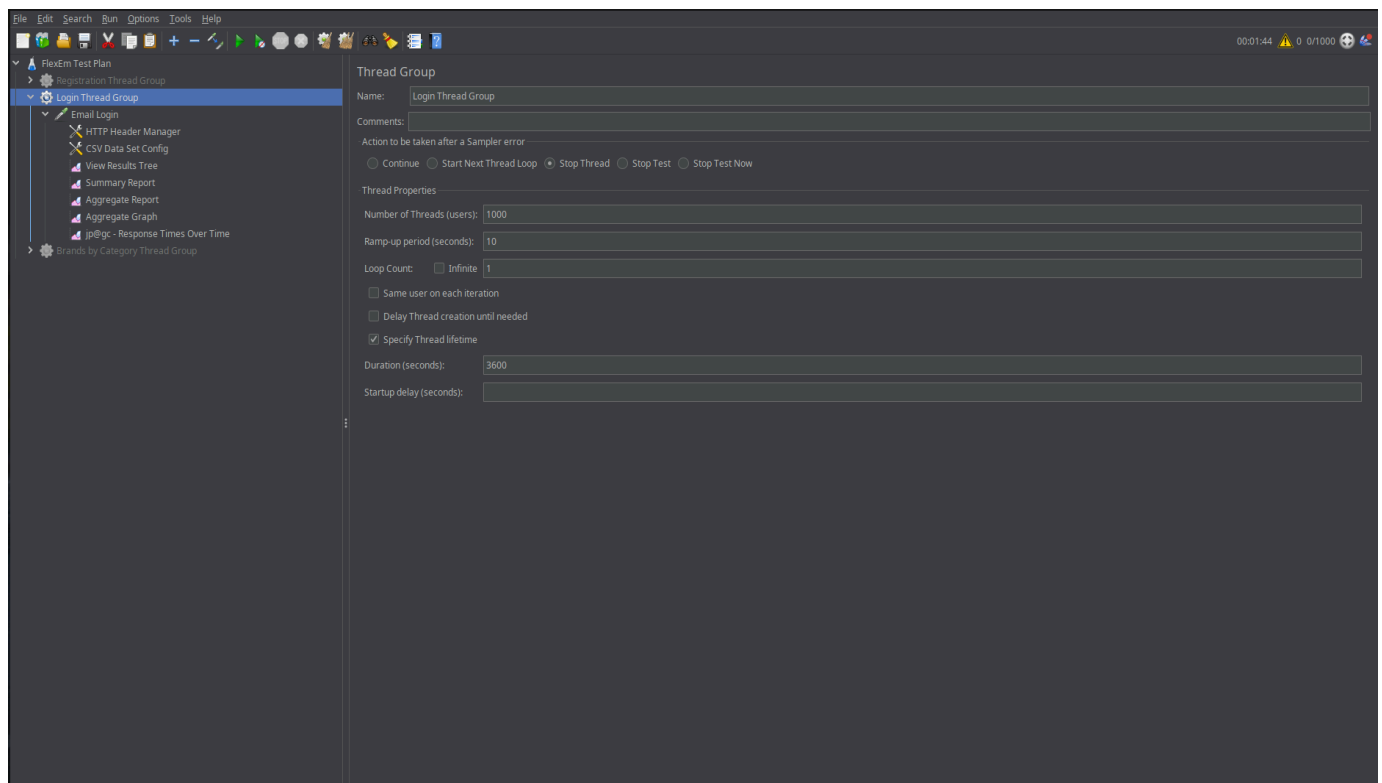
Sent KB/sec	2.66
-------------	------

2.1.4 Summary Report

The summary is the one kind of dashboard perspective on the test while the test is pursued and it has wrapped up. The Summary Report will show up when the test begins to gather Information

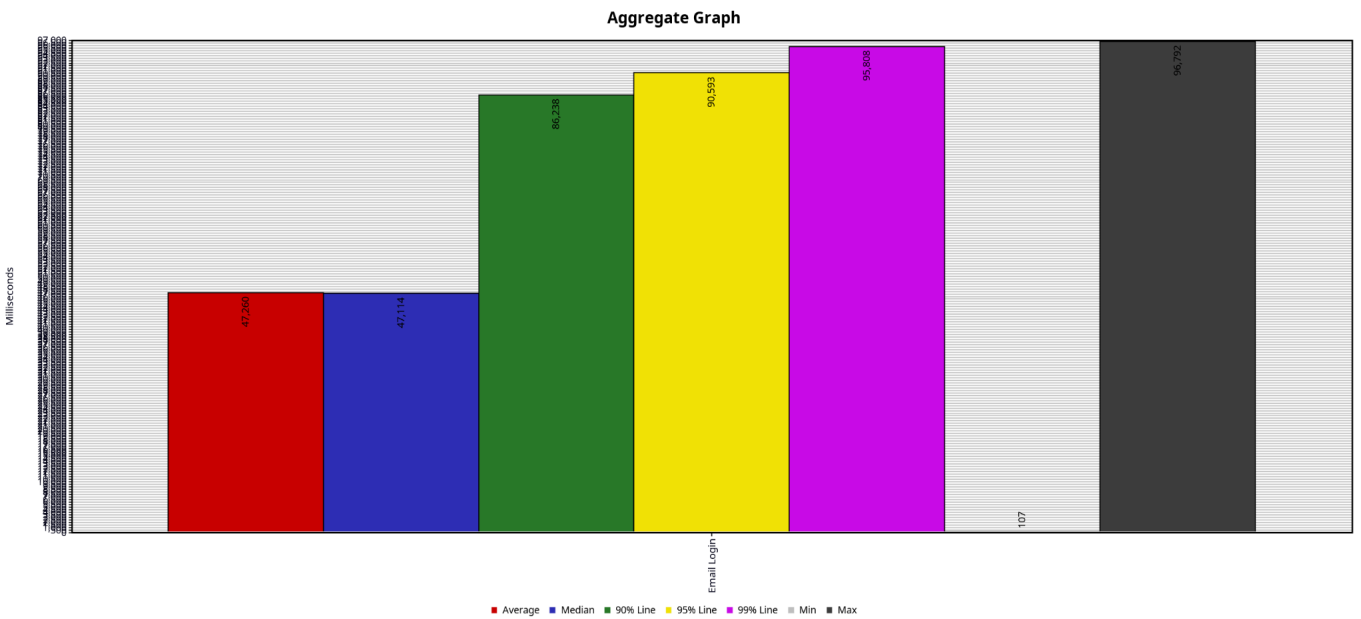
Label	Sample	Average	Min	Max	Std.Dev	Error%	Throughput	Received KB/sec	Sent KB/sec	Avg. Bytes
User Login	1000	47260	107	96792	27436.62	0.000	9.58754	9.84	2.66	1051.0

2.1.5 Configuration:



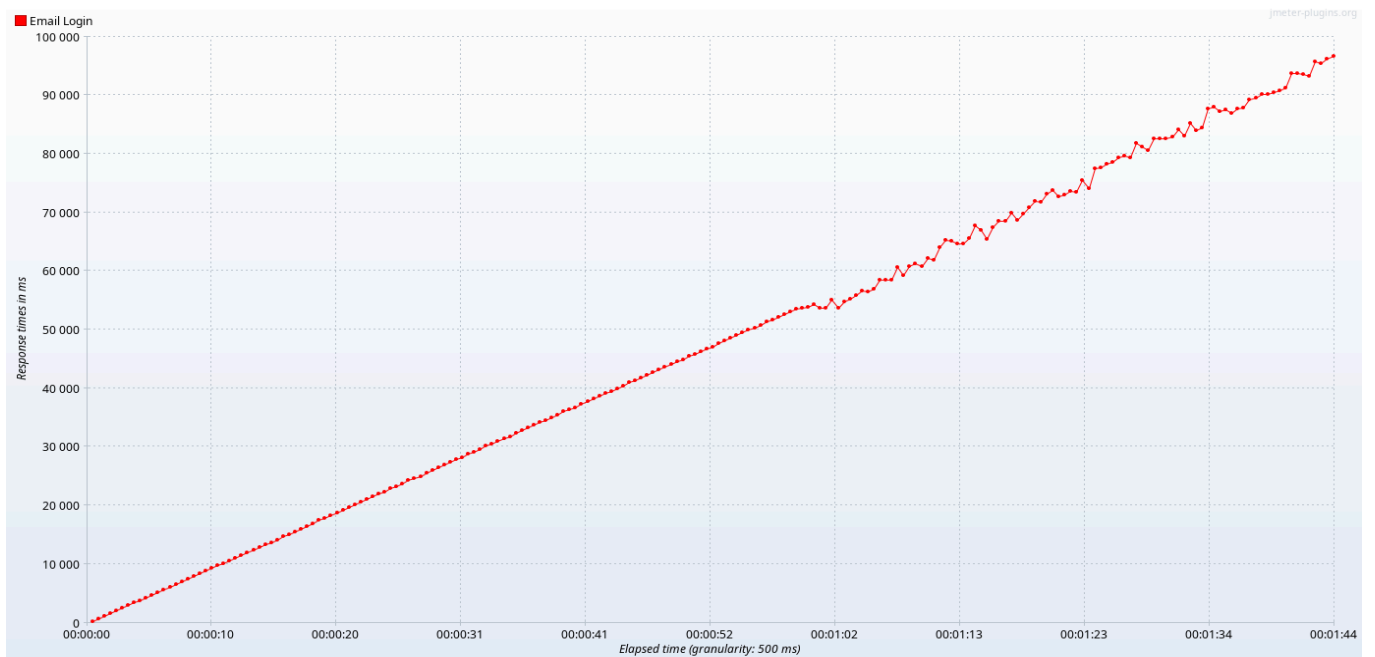
2.1.5 Output Graph

Aggregate Graph:



2.1.6 Response Time Over Time

Response Time Over Time Graph:



Test Under Sixty (60) MINUTES

Flow -> User Login, Brands By Category, Brands & Offers API:

3.1 Test with 09 RPS and 1000 User

3.1.1 Execution Summary

Following is the summary of the load test performed with 09 RPS and 1000 User under 60 minutes:

Total Requests	4000	Error/Request	0/4000
Response Time(Avg)	14541 ms	Response Time(Min)	3 ms

3.1.2 Observation Summary

- No fault found till 55 mins
- No error appeared after 60 minutes
- The system is working correctly as we've achieved our anticipated 09 TPS

3.1.3 Details

The following contains the activity report during the load test & also the performance details.

Estimated User Capacity	1000
Maximized Users Analyzed	1000
Start Time	01/06/2025 05:53:08 PM
End Time	10/06/2025 06:53:05 PM
Duration	60 min
Total Hits	4000
Total Failed Hits	0
RPS	36.36
Received KB/sec	46.06

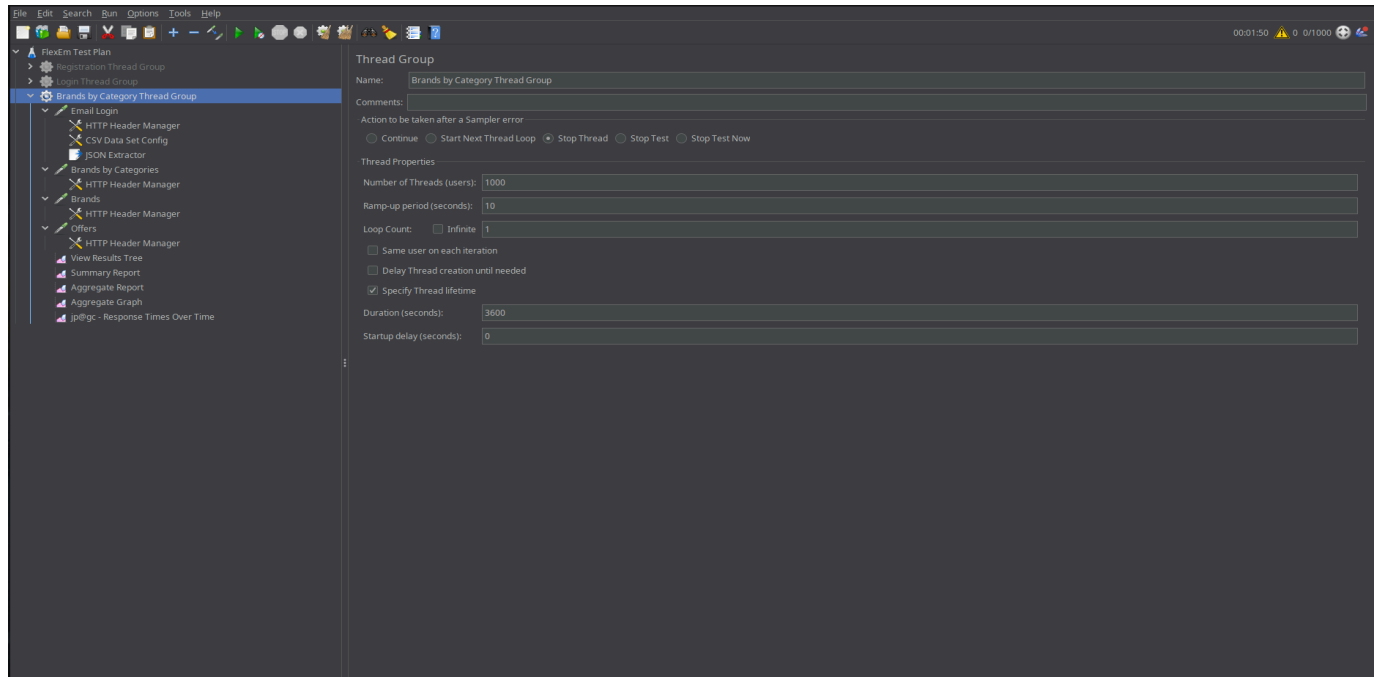
Sent KB/sec	14.68
-------------	-------

3.1.4 Summary Report

The summary is the one kind of dashboard perspective on the test while the test is pursued and it has wrapped up. The Summary Report will show up when the test begins to gather Information

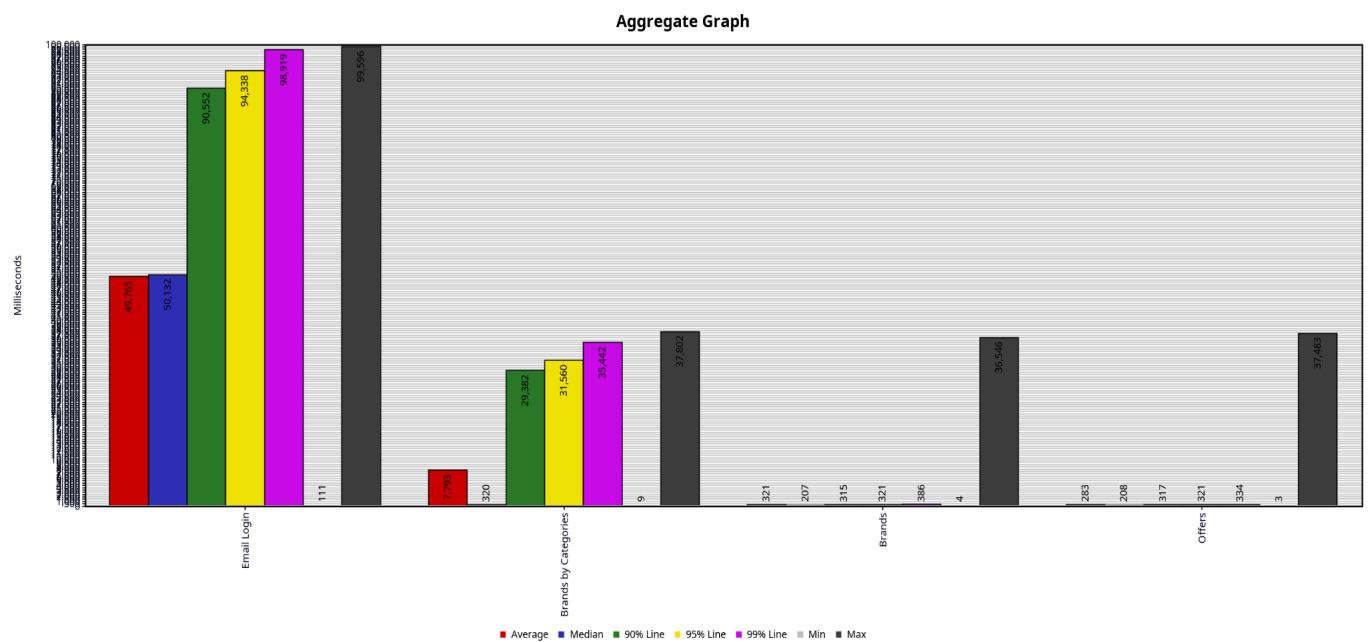
Label	Sample	Average	Min	Max	Std.Dev	Error %	Through put	Received KB/sec	Sent KB/sec	Avg. Bytes
User Login	1000	49765	111	99596	28546.97	0.000	9.23	9.48	2.56	1051.0
Brands by Category	1000	7793	9	37802	11481.88	0.000	9.10	8.60	4.08	967.0
Brands	1000	321	4	36546	2553.55	0.000	9.12	16.23	4.06	1821.0
Offers	1000	283	3	37483	2258.22	0.000	9.15	12.06	4.07	1349.0
Total	4000	14541	3	99596	25739.88	0.000	36.36827	46.06	14.68	1297.0

3.1.5 Configuration:



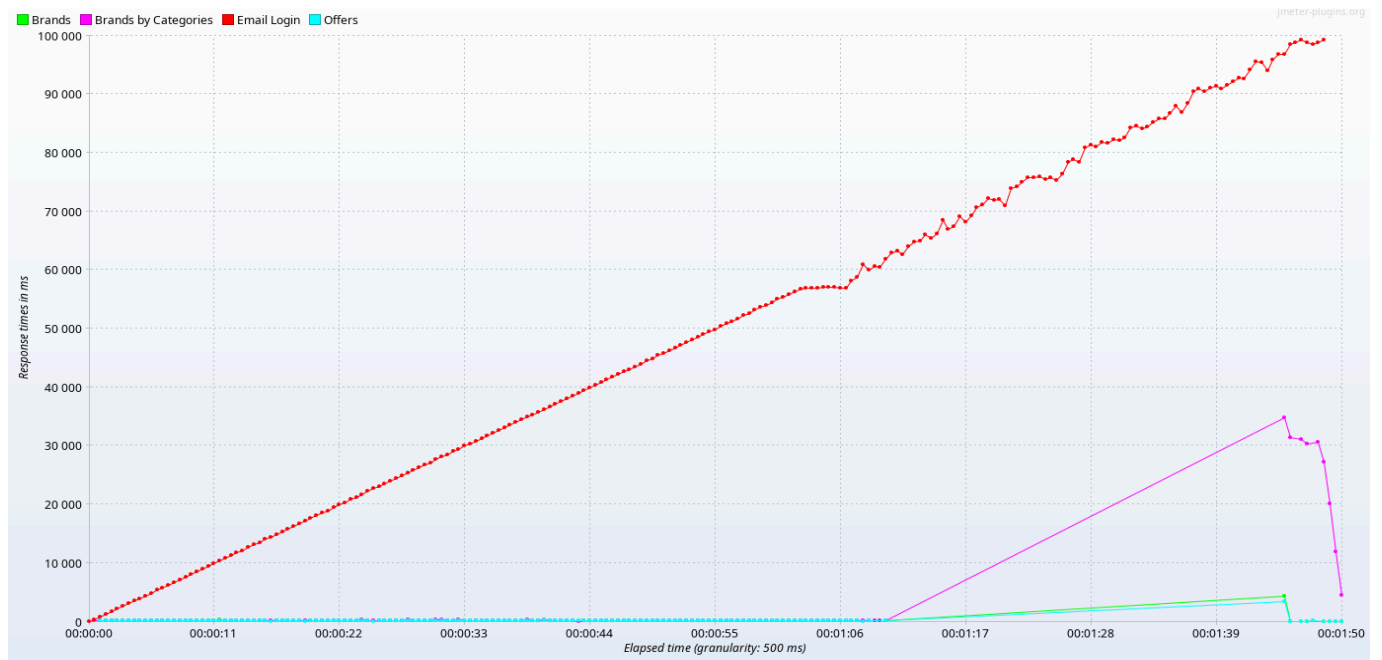
3.1.6 Output Graph

Aggregate Graph:



3.1.6 Response Time Over Time

Response Time Over Time Graph:



Test Under Sixty (60) MINUTES

Flow -> User Login, Brands By Category, Brands, Offers, Create Claim API:

4.1 Test with 09 RPS and 900 User

4.1.1 Execution Summary

Following is the summary of the load test performed with 09 RPS and 900 User under 60 minutes:

Total Requests	4500	Error/Request	8/4500
Response Time(Avg)	12407 ms	Response Time(Min)	4 ms

4.1.2 Observation Summary

- No fault found till 50 mins
- For 1000 users the server can't respond after 55 minutes & the Jmeter connection is closed. Also, there are some error appears in the Claim Create API because of a Jmeter internal configuration problem, which is a known error. That's why we've taken 900 users for this test.
- The system is working correctly, but we couldn't achieve our anticipated 09 TPS in the User Login, Brands by Category, Brands, Offers and Claim Create API together. We've achieved 08 TPS from Brands by Category, Brands, Offers APIs & achieved 09 TPS from Claim Create API.

4.1.3 Details

The following contains the activity report during the load test & also the performance details.

Estimated User Capacity	900
Maximized Users Analyzed	900
Start Time	01/06/2025 06:53:08 PM
End Time	01/06/2025 07:53:05 PM
Duration	60 min
Total Hits	4500

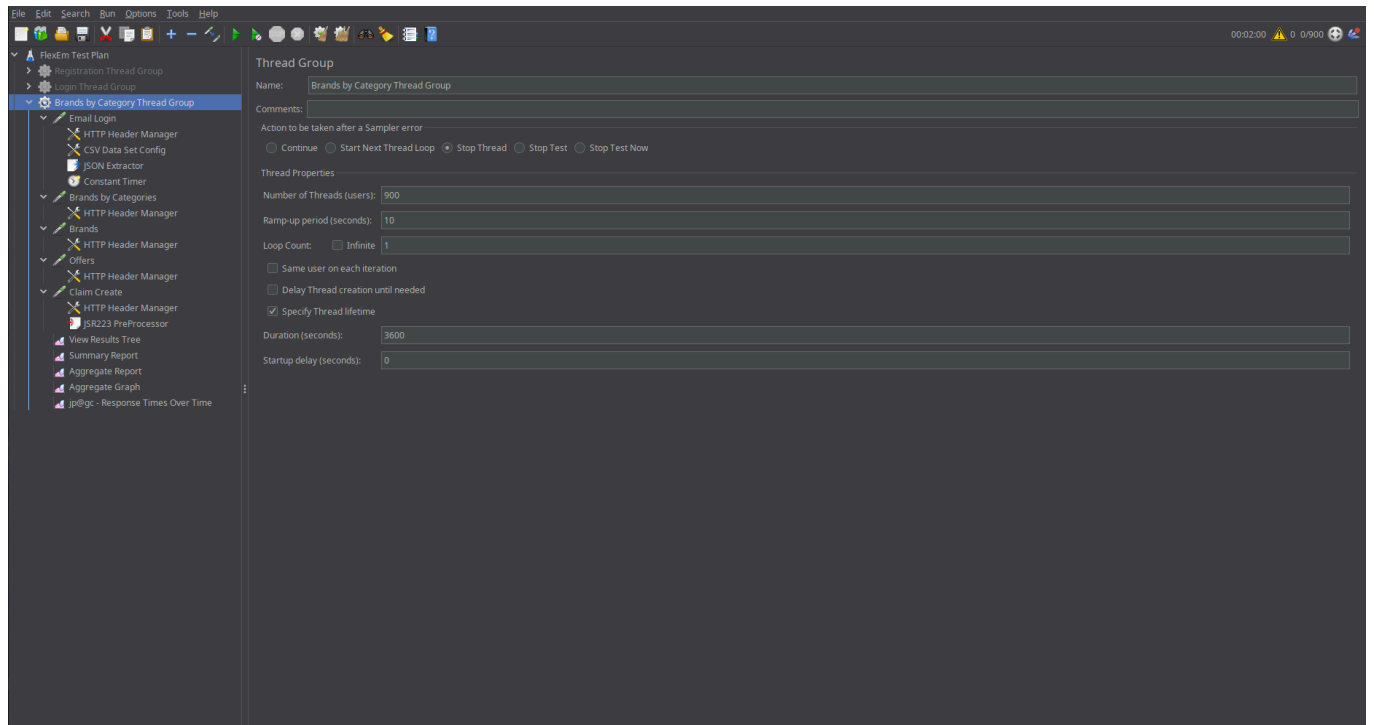
Total Failed Hits	8
RPS	44.72
Received KB/sec	47.93
Sent KB/sec	27.13

4.1.4 Summary Report

The summary is the one kind of dashboard perspective on the test while the test is pursued and it has wrapped up. The Summary Report will show up when the test begins to gather Information

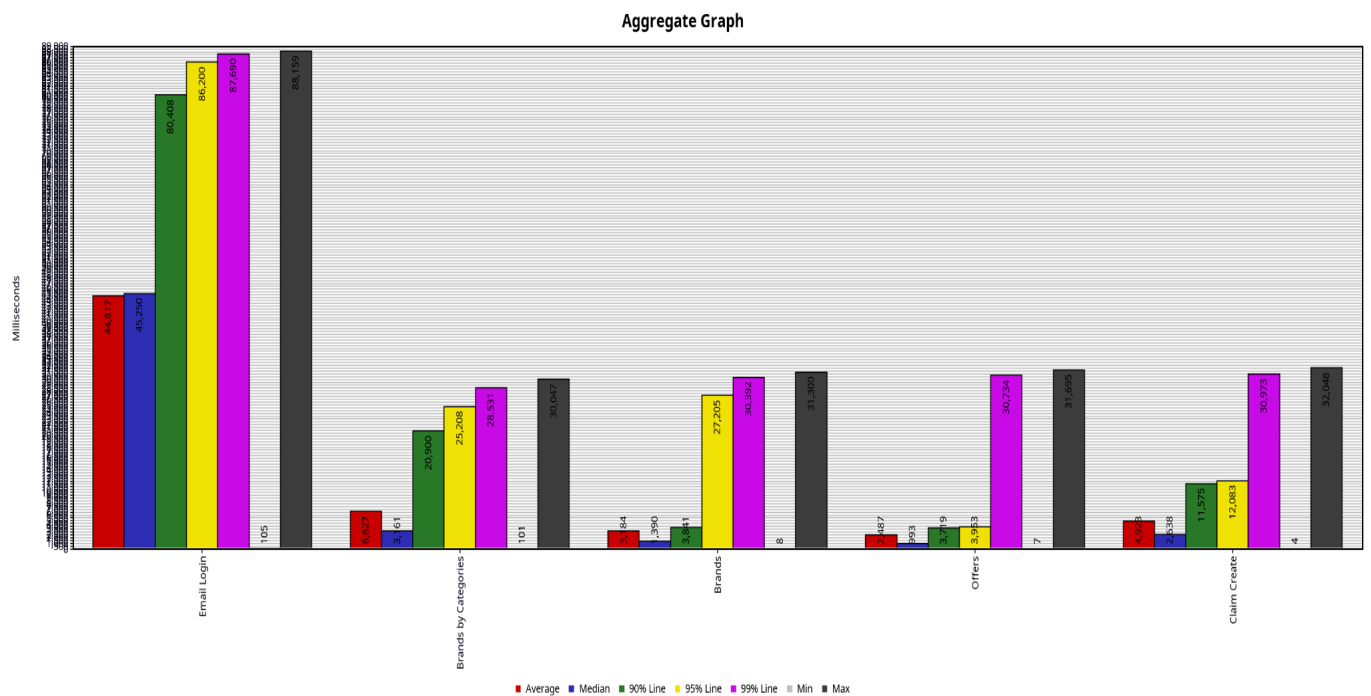
Label	Sample	Average	Min	Max	Std.Dev	Error %	Through put	Received KB/sec	Sent KB/sec	Avg. Bytes
User Login	900	44817	105	88159	25646.59	0.000	9.23	9.51	2.56	1054.3
Brands by Category	900	6627	101	30047	7837.4	0.000	8.95	8.46	4.02	967.0
Brands	900	3184	8	31300	6811.65	0.000	8.96	15.94	3.98	1821.0
Offers	900	2487	7	31695	5832.36	0.000	8.97	11.82	3.99	1349.0
Claim Create	900	4923	4	32048	6506.58	44.4	9.01	2.60	12.79	295.0
Total	4500	12407	88159	99596	20809.86	8.9	44.72583	47.93	27.13	1097.3

4.1.5 Configuration:



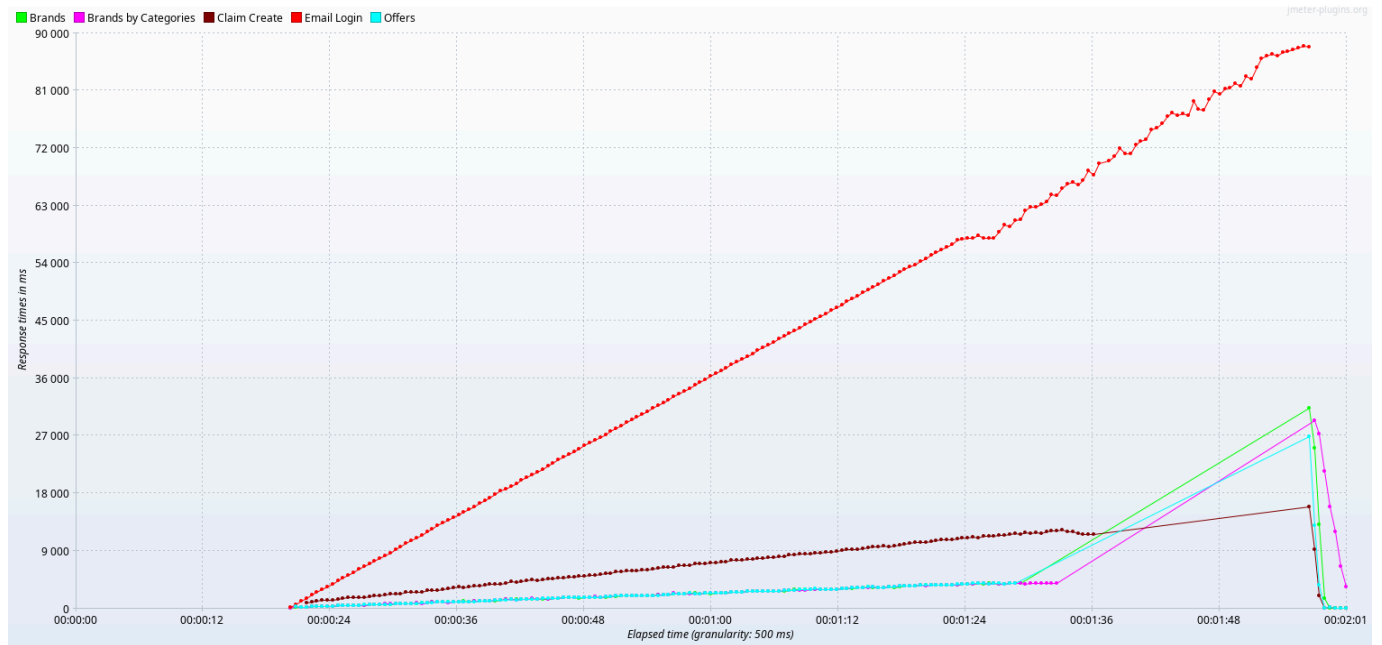
4.1.6 Output Graph

Aggregate Graph:



4.1.7 Response Time Over Time

Response Time Over Time Graph:



Test Under Sixty (60) MINUTES

Flow -> User Login, Brands By Category, Brands, Offers, Create Claim, Get Claim API:

5.1 Test with 09 RPS and 800 User

5.1.1 Execution Summary

Following is the summary of the load test performed with 09 RPS and 900 User under 60 minutes:

Total Requests	4500	Error/Request	6/4500
Response Time(Avg)	10441 ms	Response Time(Min)	3 ms

5.1.2 Observation Summary

- No fault found till 50 mins
- For 1000 & 900 users, the server can't respond after 55 minutes & the Jmeter connection is closed. Also, there are some error appears in the Claim Create API because of a Jmeter internal configuration problem, which is a known error. That's why we've taken 800 users for this test.
- The system is working correctly, but we couldn't achieve our anticipated 09 TPS from the combination of Brands By Category, Brands, Offers, Create Claim, Get Claim these APIs. We've achieved 08 TPS from Brands by Category, Brands, Offers APIs & achieved 09 TPS from User Login API.

5.1.3 Details

The following contains the activity report during the load test & also the performance details.

Estimated User Capacity	800
Maximized Users Analyzed	800
Start Time	02/06/2025 10:53:08 AM
End Time	02/06/2025 11:53:05 AM
Duration	60 min
Total Hits	4500

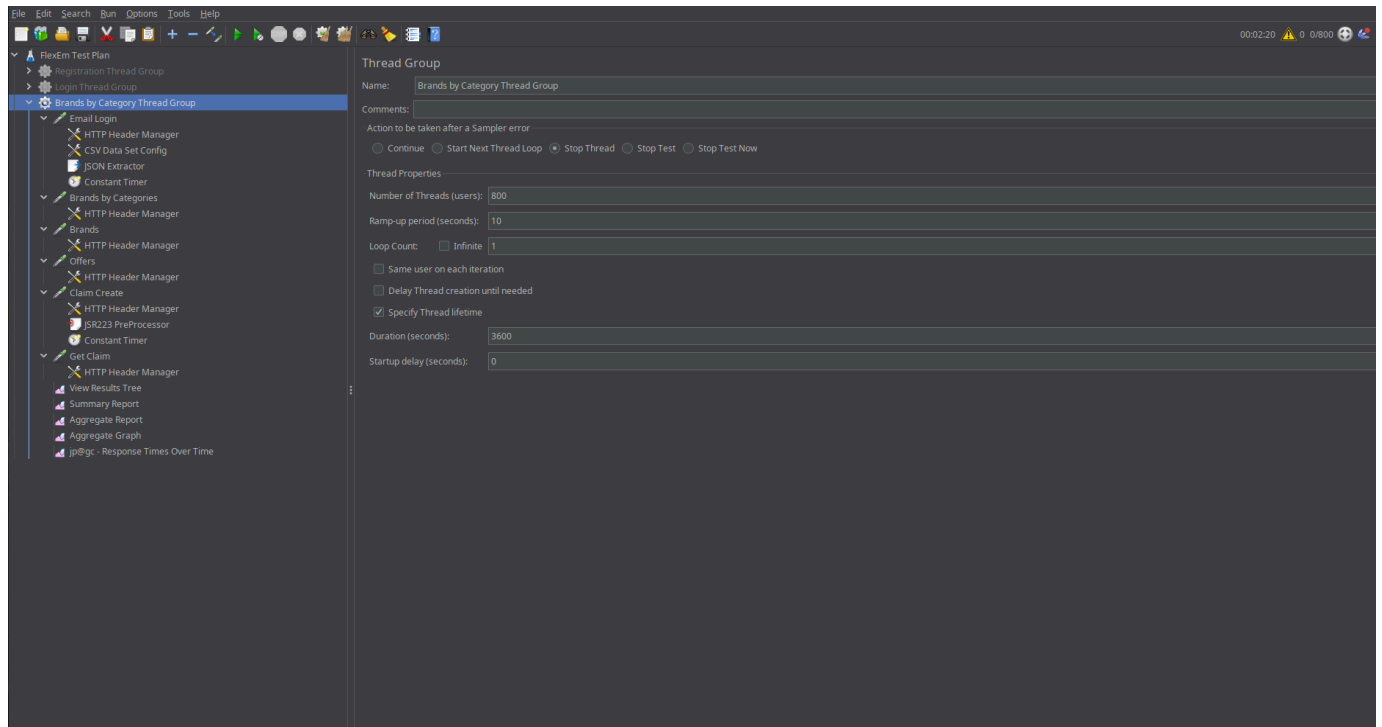
Total Failed Hits	6
RPS	37.48
Received KB/sec	45.10
Sent KB/sec	22.07

5.1.4 Summary Report

The summary is the one kind of dashboard perspective on the test while the test is pursued and it has wrapped up. The Summary Report will show up when the test begins to gather Information

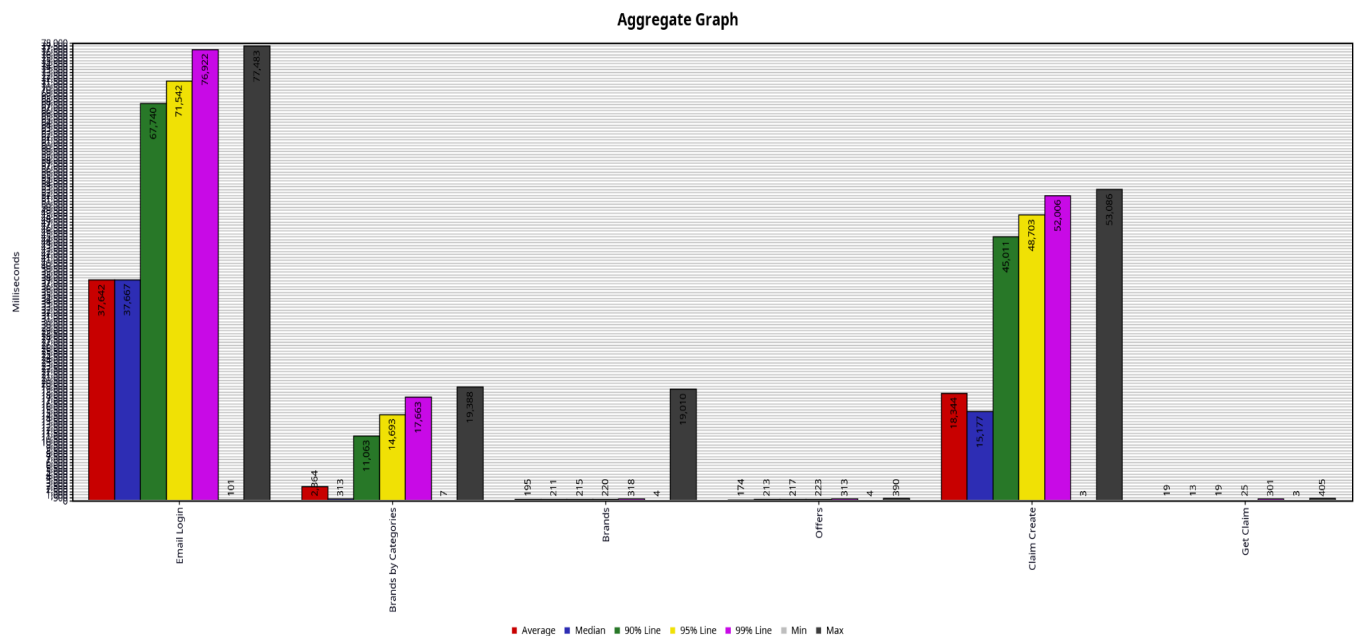
Label	Sample	Average	Min	Max	Std.Dev	Error %	Through put	Received KB/sec	Sent KB/sec	Avg. Bytes
User Login	800	37642	101	77483	21690.93	0.000	9.32	9.61	2.59	1054.8
Brands by Category	800	2364	7	19388	4639.42	0.000	8.89	8.40	3.99	967.0
Brands	800	195	4	19010	671.04	0.000	8.90	15.84	3.96	1821.0
Offers	800	174	4	390	85.81	0.000	8.92	11.76	3.97	1349.0
Claim Create	800	18344	3	53086	17661.91	37.5	8.92	2.53	12.67	290.6
Get Claim	500	19	3	405	39.45	0.000	62.07	140.43	27.58	2316.7
Total	4500	10441	3	77483	18620.3	6.7	37.48626	45.1	22.07	1232.1

5.1.5 Configuration:



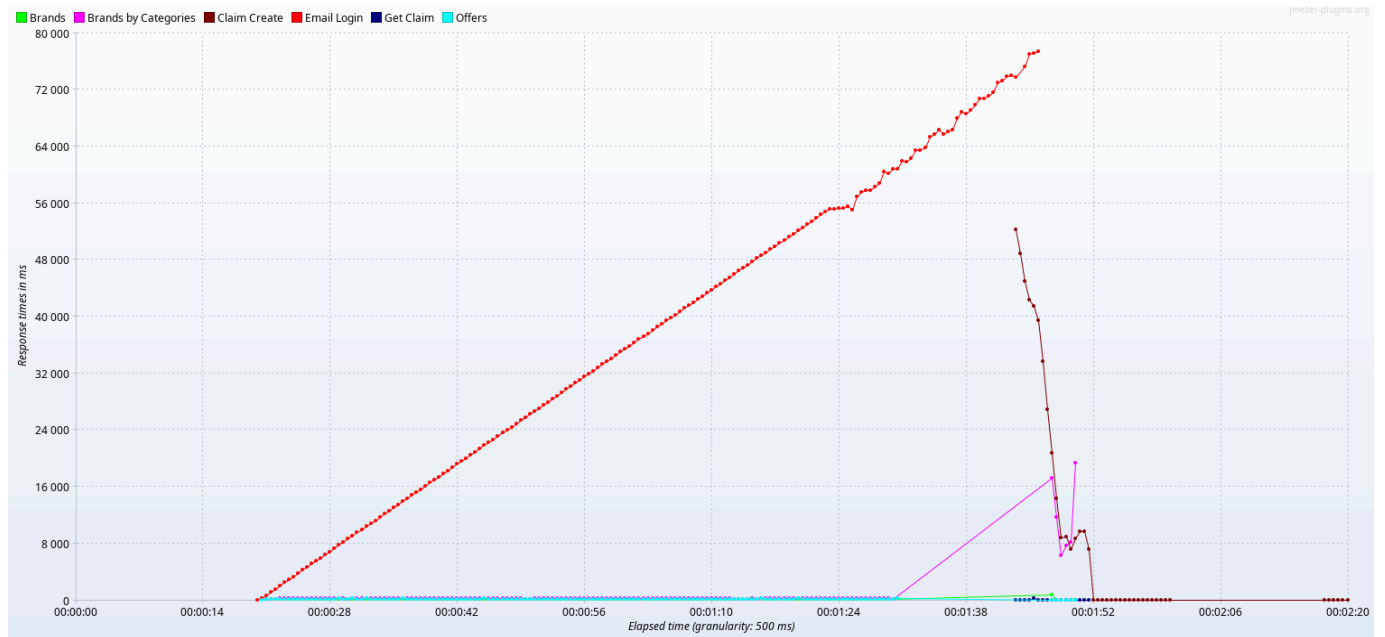
5.1.6 Output Graph

Aggregate Graph:



5.1.7 Response Time Over Time

Response Time Over Time Graph:



Test Under Eight Hours (480) MINUTES